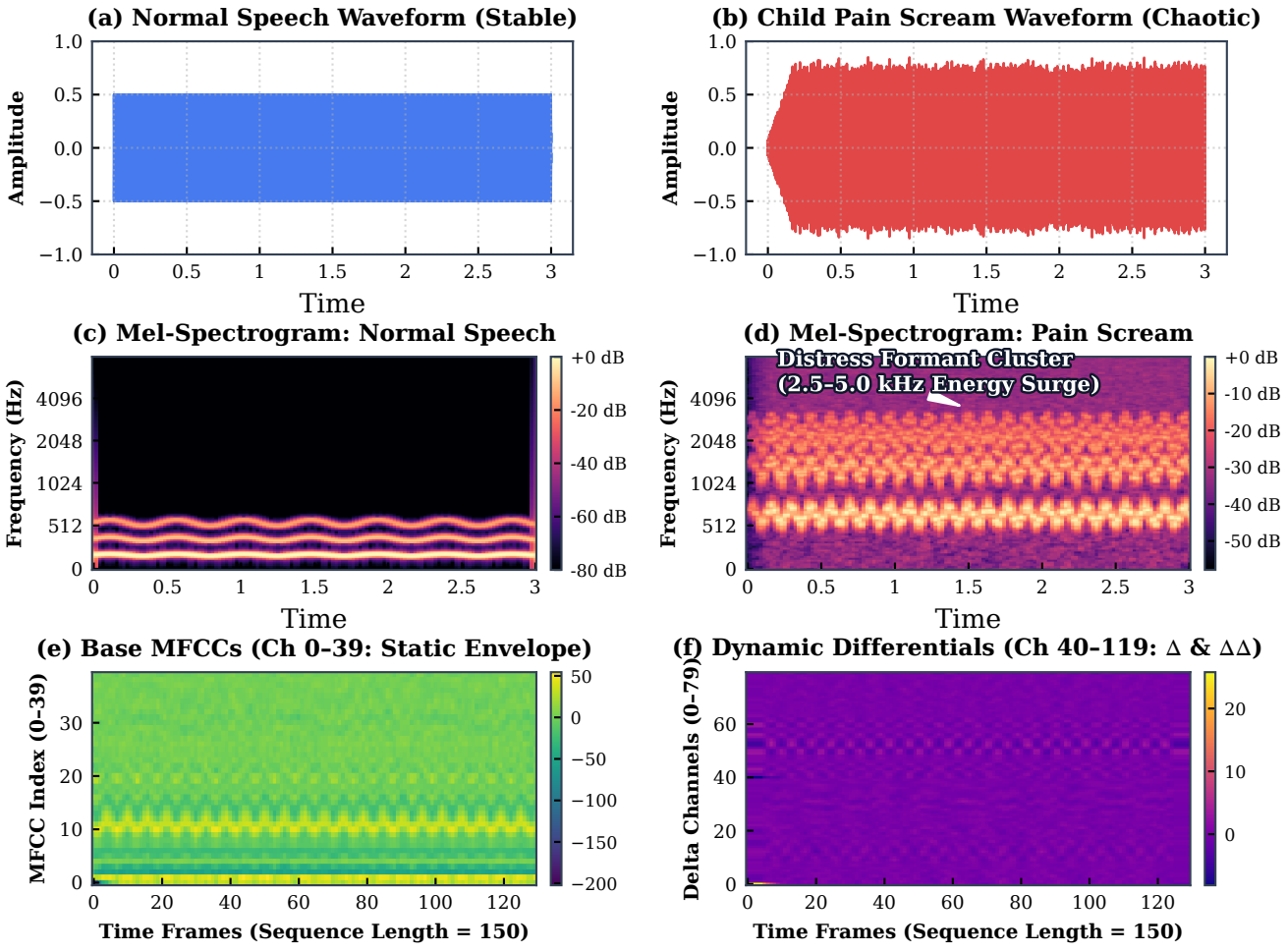

(a) Normal Speech Waveform (Stable)
Amplitude
1.0
0.5
0.0
−0.5
−1.0
0
0.5
1
1.5
2
2.5
3
Time
(b) Child Pain Scream Waveform (Chaotic)
Amplitude
Time
(c) Mel-Spectrogram: Normal Speech
Frequency (Hz)
4096
2048
1024
512
0
+0 dB
-20 dB
-40 dB
-60 dB
-80 dB
Time
(d) Mel-Spectrogram: Pain Scream
Distress Formant Cluster
(2.5–5.0 kHz Energy Surge)
Frequency (Hz)
+0 dB
-10 dB
-20 dB
-30 dB
-40 dB
-50 dB
Time
(e) Base MFCCs (Ch 0–39: Static Envelope)
MFCC Index (0–39)
30
20
10
0
50
0
−50
−100
−150
−200
0
20
40
60
80
100
120
Time Frames (Sequence Length = 150)
(f) Dynamic Differentials (Ch 40–119: Δ & ΔΔ)
Delta Channels (0–79)
60
40
20
0
20
10
0
Time Frames (Sequence Length = 150)